NETFLIX RECOMMENDATION SYSTEM

PROBLEM STATEMENT & OBJECTIVE

GOAL

BUILD A PERSONALIZED RECOMMENDATION SYSTEM USING THE NETFLIX PRIZE DATASET CAPABLE OF:

- Learning user preferences from historical ratings
- Predicting unseen movie ratings
- Generating Top-K personalized recommendations
- Evaluating recommendation quality using RMSE and MAP@10

Dataset Challenges

- 100M+ RATINGS
- 480K+ USERS
- 17K+ MOVIES
- EXTREMELY SPARSE USER-ITEM MATRIX
- COLD-START USERS AND ITEMS
- TEMPORAL EVOLUTION OF PREFERENCES

Project Approach Compare:

- Item-Based Collaborative Filtering
- Matrix Factorization (SVD)
- SVD++

under multiple evaluation strategies.

DATA UNDERSTANDING & EDA

DATASET USED

ORIGINAL NETFLIX PRIZE DATASET

Metric	Value
Ratings	100,480,507
Users	480,189
Movies	17,770

MODELING DATASET

DUE TO COMPUTATIONAL CONSTRAINTS:

- SAMPLED 2,000,000 RATINGS
- 358,375 USERS
- 17,324 MOVIES

1999–2005 TIME PERIOD

Key EDA Findings

Rating Distribution

- Mean Rating = 3.61
- Median Rating = 4
- Majority ratings between 3 and 5

User Activity

- Most users have very few ratings
- Few highly active users dominate interactions

Movie Popularity

- Strong long-taildistribution
- Small set of movies receives most ratings

IMPORTANT INSIGHT
ALTHOUGH ONLY 2M RATINGS WERE USED, THE SAMPLE RETAINED:

17,324 MOVIES OUT OF 17,770
≈97.5% MOVIE COVERAGE

MAKING IT HIGHLY REPRESENTATIVE OF THE ORIGINAL CATALOG.

EVALUATION STRATEGY & SPLIT ANALYSIS

Why Multiple Train-Test Splits?

RANDOM SPLITTING OFTEN LEAKS FUTURE INFORMATION. THEREFORE, THREE EVALUATION STRATEGIES WERE EXPLORED.

1. RANDOM SPLIT

80% TRAIN
20% TEST
PROS:
- EASY BENCHMARK

CONS:
- UNREALISTIC FUTURE PREDICTION

2. GLOBAL TEMPORAL SPLIT

PAST RATINGS → TRAIN
FUTURE RATINGS → TEST

MORE REALISTIC FOR PRODUCTION SYSTEMS.

3. PER-USER TEMPORAL SPLIT

FOR EVERY USER:

OLD RATINGS → TRAIN
RECENT RATINGS → TEST

MOST REALISTIC EVALUATION.

Important Observation

Cold-start analysis revealed:

Unseen Test Users = 57,336

Cold User Ratings ≈ 48%

showing why recommendation is difficult in real-world settings.

RECOMMENDATION MODELS

MODEL 1: ITEM-BASED COLLABORATIVE FILTERING

IDEA

RECOMMEND MOVIES SIMILAR TO THOSE ALREADY LIKED BY A USER.

METHOD

USER-ITEM RATING MATRIX
COSINE SIMILARITY
NEIGHBORHOOD-BASED PREDICTION

ADVANTAGES

SIMPLE
EXPLAINABLE

LIMITATIONS

SPARSE DATA PROBLEM
POOR SCALABILITY

MODEL 2: SINGULAR VALUE DECOMPOSITION (SVD)

IDEA

LEARN HIDDEN LATENT FACTORS REPRESENTING:
USER PREFERENCES
MOVIE CHARACTERISTICS

BENEFITS

HANDLES SPARSITY BETTER
CAPTURES HIDDEN RELATIONSHIPS

MODEL 3: SVD++

EXTENSION OF SVD USING IMPLICIT FEEDBACK FROM USER INTERACTIONS.

EXPERIMENTAL RESULTS (RMSE)

Rating Prediction Performance

- ModelRMSE
- Item-CF Random - 1.088
- SVD Random - 0.985
- Item-CF Temporal - 1.125
- SVD Temporal - 1.034
- Item-CF Per User Temporal - 1.154
- SVD Per User Temporal - 0.973
- Tuned SVD- 970
- SVD++ - 0.979

Hyperparameters of Best Model

n_factors = 100
n_epochs = 50
learning_rate = 0.003
regularization = 0.05

Key Findings

- SVD CONSISTENTLY OUTPERFORMED ITEM-CF.
- TEMPORAL EVALUATION IS SIGNIFICANTLY HARDER THAN RANDOM EVALUATION.
- TUNED SVD ACHIEVED THE BEST PREDICTION ACCURACY.

Recommendation Quality & Top-K Recommendations

Top-K Recommendation Generation

Recommendations generated using the best performing model (Tuned SVD).

Example Recommendations

Movie	Predicted Rating
The West Wing: Season 3	4.75
Lord of the Rings: Return of the King	4.70
Spirited Away	4.69
The Sopranos: Season 3	4.67
Lost: Season 1	4.64

Observation

The model successfully recommends:

- **Critically acclaimed movies**
- **Popular TV series**
- **Highly rated content indicating meaningful preference learning.**

Ranking Evaluation & Cold Start Analysis

Ranking Metrics
Original Evaluation
MetricScore

Precision@10 - 0.594
Recall@10 - 0.999
MAP@10 - 0.644

Final Evaluation
MetricScore

Precision@10 - 0.437
Recall@10 - 0.898
MAP@10 - 0.597

Conclusions

Best Model
Tuned SVD
RMSE = 0.970

Major Findings

- Matrix factorization outperformed collaborative filtering.
- Temporal evaluation provided more realistic performance estimates.
- Recommendation quality remained strong under stricter evaluation.
- The sampled dataset preserved almost the entire movie catalog.
- Cold-start users remain challenging.

Cold Start Results
SegmentRMSE

Cold - 1.697
Light- 1.737
Medium - 1.910
Heavy - 3.211

Issue Identified

Many users had very few test interactions.
To obtain a more realistic estimate:
Only users with ⩾10 test ratings were evaluated.

Future improvements

- Alternating Least Squares (ALS)
- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Content-Based Features
- Full Netflix Dataset Training

Final Takeaway

The project demonstrates that latent-factor models such as SVD can effectively capture user preferences and generate meaningful recommendations at scale while balancing predictive accuracy and ranking quality.

Insight

Cold-start remains one of the most significant challenges for recommendation systems.

Thank you

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